

Wired, December 2004

I Spy Your Cell Phone

AND WE THOUGHT they would at least leave Cell phones alone. Annalee Newitz of *Wired* tells us how the tech gadget you use the most is also the most vulnerable of them all to virus attacks and spyware.

It's a beautiful afternoon in Shepherd's Bush, a bustling neighborhood on the outskirts of London, and Adam Laurie is feeling peckish. Heading out of the office, he's about to pick up more than a sandwich. As he walks, he'll be probing every cell phone that comes within range of a hidden antenna he has connected to the laptop in his bag. We stroll past a park near the Tube station, then wander into a supermarket. Laurie contemplates which sort of crisps to buy while his laptop quietly scans the 2.4-GHz frequency range used by Bluetooth



devices, probing the cell phones nestled in other shoppers' pockets and purses.

Laurie, 42, the CSO of boutique security firm the Bunker, isn't going to mess with anyone's phone, although he could: With just a few tweaks to the scanning program his computer is running, Laurie could be crashing cell phones all around him, cutting a little swath of telecommunications destruction down the deli aisle. But today Laurie is just gathering data. We are counting how many phones he can hack using Bluetooth, a wireless protocol for syncing cell phones with headsets, computers, and other devices.

We review the results of the expedition in a nearby pub. In the 17 minutes we wandered around, Laurie's computer picked up signals from 39 phones. He peers at his monitor for a while. "It takes only 15 seconds to suck down somebody's address book, so we could have had a lot of those," he says at last. "And at least five of these phones were vulnerable to an attack."

The "attack" Laurie mentions so casually could mean almost anything—a person using another person's cell to make long distance calls or changing every phone number in his address book or even bugging his conversations. There are, he says, "a whole range of new powers" available to the intrepid phone marauder, including nasty viral attacks. A benign Bluetooth worm has already been discovered circulating in Singapore, and Laurie thinks future variants could be something really scary. Especially vulnerable are Europeans who use their mobile phone to make micropayments—small purchases that show up as charges on cell phone bills. A malicious virus maker bent on a get-rich-quick scheme could take advantage of this feature by issuing "reverse SMS" orders.

Bluetooth security has become a pressing issue in Europe, where the technology is ubiquitous. The problem will migrate to American shores as the protocol catches on here, too. But in the long run, Bluetooth vulnerabilities are manageable: Handset manufacturers can rewrite faulty implementations, and cell phone users will learn to be more careful.

To understand in detail how your cell phone could be attacked any moment, visit <http://snipurl.com/digitphreak>

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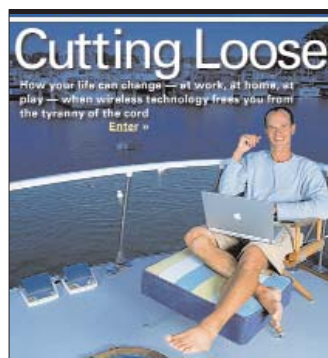
The Spokane (Wireless) Word

SOME CITIES ARE born great, while some have greatness thrust upon them. Spokane, a non-descript city in north-western United States is now the wireless capital of the world. *Time* magazine's Lev Grossman gives us a walkthrough of this previously-unheard of city

I rolled into Spokane, Wash., around 6 p.m. on a Tuesday. Spokane is one of those sleepy cities bursting with small-town pride—its residents will be glad to inform you, for example, that it's the smallest city ever to be host of a World's Fair—but it's a pretty quiet place on a Tuesday night. You can look both ways before you cross the street if you really want to, but it's just a formality. I sat on a park bench. A dude on the corner played the saxophone. Some punks on dirt bikes made fun of me. The silence was eerie. Zombie-movie eerie.

But there's a lot more going on here than meets the eye. Spokane is actually a radical experiment in urban wireless technology, a live-in laboratory where city-employed nerds are

crash-testing the wireless technopia of the future. All of downtown Spokane, including the park that I was sitting in, is a massive wi-fi hot spot, a whole neighborhood enveloped in an invisible field of high-volume Internet access that covers 100 city blocks. The same way some libraries and coffeehouses offer wireless Internet access, all of downtown Spokane is a wireless



surfing zone. If I'd had a laptop with a wi-fi card, or even a Web-enabled PDA or cell phone, I could have surfed the Web right from that park bench for free.

Sadly, I didn't. That would have shown up those dirt-bike punks.

Spokane is by no means the only project of its kind. It's easy to imagine that by the end of the decade most U.S. cities will exist beneath an invisible dome of wi-fi—"city clouds," in the jargon of the industry. Rio Rancho, N.M., has one, though not on the scale of Spokane's; ditto for Grand Haven, Mich. (see sidebar), as well as Lafayette, Louis and Cerritos in California. And bigger players are moving in all the time. Cook County, Ill., is planning a massive 940-sq.-mi. cloud that would light up all of Chicago. Philadelphia announced a humongous hot zone of its own in September. Los Angeles and New York City are soliciting bids from wireless contractors. This stuff is just too cheap and too useful not to have. It doesn't even stop at the city limits. Out in the sticks, where there are no skyscrapers to get in the way of a wi-fi signal, wireless is even bigger. There's a hot spot in rural Walla Walla County, Wash., that runs 3,700 sq. mi.

I was in Spokane to meet the people behind its audacious experiment, principally a guy named Don Stalter, ceo of Vivato, the high-tech start-up that supplies the technology to make it possible. Stalter didn't found the company; it began with a Hewlett-Packard engineer named Skip Crilly, who lived in the hills outside Spokane and couldn't get anybody to run a high-speed line to his house.

Read more at <http://snipurl.com/digitwireless>